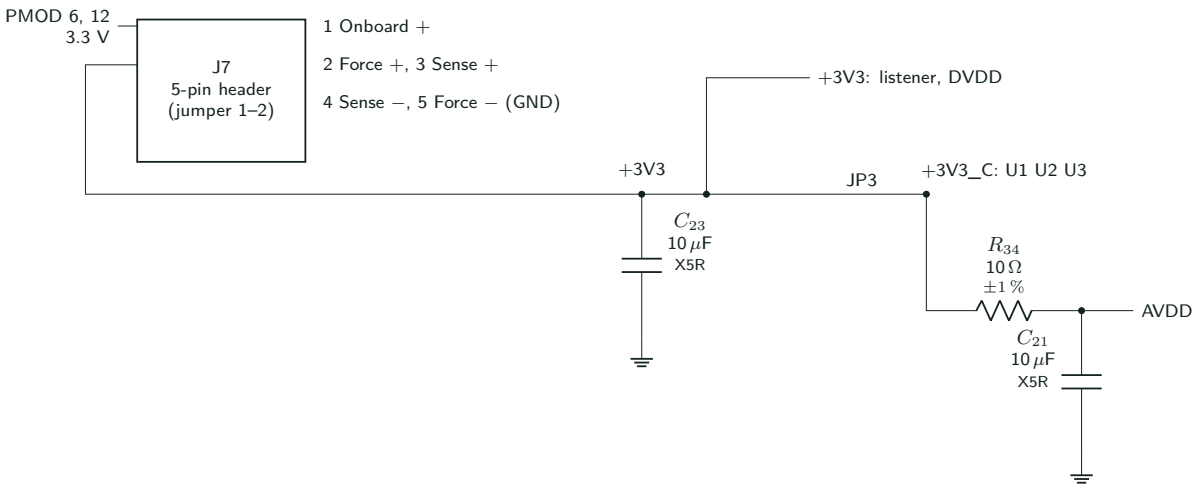
# Slide 10: power and rails

3.3 V from the PMOD alone, split by current links

ADS7057 test board v3 — study notes

## 1 The power tree



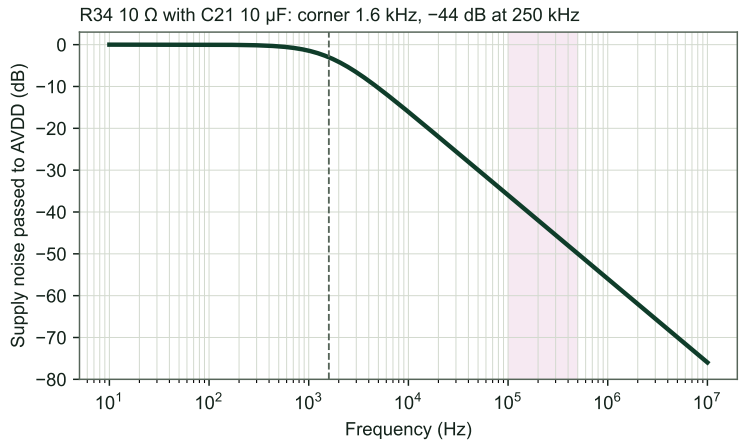

PMOD pins 6 and 12 bring 3.3 V. There is no regulator, no diode and no USB. The 5-pin header J7 is the power entry: 1 Onboard + (PMOD 3.3 V), 2 Force +, 3 Sense +, 4 Sense −, 5 Force −. A jumper from 1 to 2 powers the board from the ZedBoard. For a source-measure unit remove the jumper and connect Force to 2 and 5, Sense to 3 and 4: the SMU powers the board and reads its total current. JP3 (closed) joins the capture rail +3V3_C (U1, U2, U3 and AVDD) to +3V3. Open it to switch or measure the chain alone. The listener and DVDD sit on +3V3.

## 2 AVDD is the reference

The ADS7057 uses AVDD as its reference, so noise on AVDD is noise in the code. AVDD comes from the rail through R34 (10 Ω) and C21 (10 µF). First-order RC:

$$f_c = \frac{1}{2\pi R_{34} C_{21}} = \frac{1}{2\pi \cdot 10\,\Omega \cdot 10\,\mu\text{F}} = 1.59\,\text{kHz}$$

$$|H(250\,\text{kHz})| = \frac{1}{\sqrt{1 + (250/1.59)^2}} = \frac{1}{157} = -43.9\ \text{dB}.$$

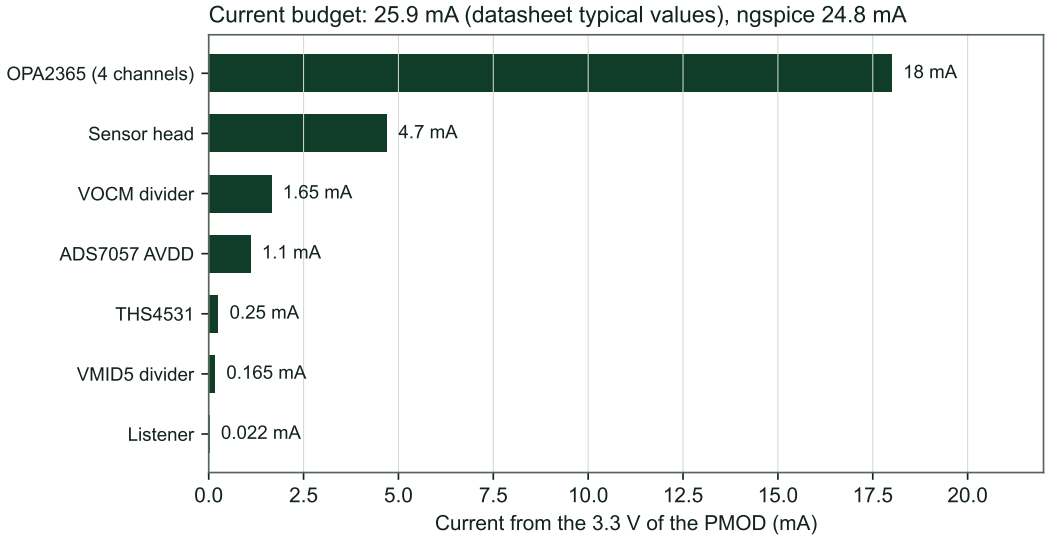



C18 (4.7 $\mu$F) and C22 (100 nF) sit at the ADC pin. The DC drop over R34 is small: the ngspice rail shows AVDD at 3.271 V against 3.299 V on +3V3.

## 3 The two mid-rails

VMID5 (the 1.65 V that the filters and the gain stage reference) is a divider R7, R8 ($2 \times 10\,\text{k}\Omega$ from +3V3_C) with C7 (1 $\mu$F), copied by U2B and filtered by R9 and C8. The divider and C7 give $\tau = (R_7 \parallel R_8)C_7 = 5\,\text{k}\Omega \times 1\,\mu\text{F} = 5\,\text{ms}$, so VMID5 needs about $5\tau = 25\,\text{ms}$ after power-up. VOCM (the driver's centre) is a second divider R14, R15 ($2 \times 1\,\text{k}\Omega$ from AVDD) with C13. It follows AVDD by ratio, as the datasheet asks (AVDD/2 ±0.1 V). ngspice: VMID5 1.649 V, VOCM 1.636 V.

## 4 Current budget




The sum of the datasheet typical values is 25.9 mA, which is 85 mW at 3.3 V. The ngspice netlist with all links closed gives 24.8 mA. The four OPA2365 channels dominate. The listener takes only 22 µA. The 1.65 mA of the VOCM divider is the cheapest saving if the board ever runs from a battery (use 10 kΩ and a larger C13).

Open item: the noise of the ZedBoard 3.3 V rail is not measured. Probe TP11 (+3V3) and TP12 (AVDD) on the first board.